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Electroluminescent device having light transmitting region of non-through-hole structure

Abstract

A display device assembly including a glass substrate and a display device disposed on the glass substrate, the display device including a first portion and a second portion having a light transmittance greater than the first portion, in which the display device includes at least one display pixel, and a lower portion of the glass substrate has a recess vertically corresponding to the second portion of the display device.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7733010	12/2009	Lee et al.	N/A	N/A
9190630	12/2014	Kim et al.	N/A	N/A
9935161	12/2017	Kim et al.	N/A	N/A
9940888	12/2017	Ryoo et al.	N/A	N/A
10033017	12/2017	Kim	N/A	N/A
10109692	12/2017	Senda et al.	N/A	N/A
10186191	12/2018	Kang et al.	N/A	N/A
10297652	12/2018	Kim et al.	N/A	N/A
10340476	12/2018	Park	N/A	N/A
10482811	12/2018	Kang et al.	N/A	N/A
10529952	12/2019	Kim et al.	N/A	N/A
10714707	12/2019	Kim et al.	N/A	N/A
10818874	12/2019	Park	N/A	N/A
10910600	12/2020	Park	N/A	N/A
10978666	12/2020	Park	N/A	N/A
11127920	12/2020	Park	N/A	N/A

11251401	12/2021	Kim et al.	N/A	N/A
11444261	12/2021	Park	N/A	N/A
11515507	12/2021	Park	N/A	N/A
11737311	12/2022	Kim et al.	N/A	N/A
2008/0117497	12/2007	Shimodaira	N/A	N/A
2009/0257105	12/2008	Xu et al.	N/A	N/A
2013/0313672	12/2012	Min	N/A	N/A
2014/0021574	12/2013	Egawa	N/A	N/A
2014/0145155	12/2013	Park	N/A	N/A
2014/0148155	12/2013	Batta et al.	N/A	N/A
2015/0060778	12/2014	Kim et al.	N/A	N/A
2016/0126494	12/2015	Jung et al.	N/A	N/A
2016/0170524	12/2015	Kim et al.	N/A	N/A
2016/0221293	12/2015	Park	N/A	N/A
2016/0233289	12/2015	Son et al.	N/A	N/A
2016/0254479	12/2015	Jeong et al.	N/A	N/A
2016/0337570	12/2015	Tan et al.	N/A	N/A
2017/0047544	12/2016	Kang et al.	N/A	N/A
2017/0053973	12/2016	Park et al.	N/A	N/A
2017/0077456	12/2016	Chung et al.	N/A	N/A
2017/0148856	12/2016	Choi et al.	N/A	N/A
2017/0154566	12/2016	Ryoo et al.	N/A	N/A
2017/0162111	12/2016	Kang et al.	N/A	N/A
2017/0162637	12/2016	Choi et al.	N/A	N/A
2017/0187934	12/2016	Kwak et al.	N/A	N/A
2017/0199400	12/2016	Mok et al.	N/A	N/A
2017/0221976	12/2016	Park et al.	N/A	N/A
2017/0237037	12/2016	Choi et al.	N/A	N/A
2017/0237038	12/2016	Kim et al.	N/A	N/A
2017/0256747	12/2016	Lee et al.	N/A	N/A
2017/0263895	12/2016	Lee et al.	N/A	N/A
2017/0277219	12/2016	Chung et al.	N/A	N/A
2017/0287992	12/2016	Kwak et al.	N/A	N/A
2017/0289324	12/2016	Yeo et al.	N/A	N/A
2017/0294502	12/2016	Ka et al.	N/A	N/A
2018/0138450	12/2017	Park et al.	N/A	N/A
2019/0214595	12/2018	Park	N/A	N/A
2020/0051497	12/2019	Kang et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
103811530	12/2013	CN	N/A
105633117	12/2015	CN	N/A
107025875	12/2016	CN	N/A
107452775	12/2016	CN	N/A
10-1325577	12/2012	KR	N/A
10-2014-0038827	12/2013	KR	N/A
10-2017-0064598	12/2016	KR	N/A
10-2017-0065059	12/2016	KR	N/A
10-2017-0095444	12/2016	KR	N/A

10-2017-0096646	12/2016	KR	N/A
10-2017-0104097	12/2016	KR	N/A
10-2017-0111827	12/2016	KR	N/A
10-2017-0113932	12/2016	KR	N/A
10-2017-0117291	12/2016	KR	N/A
10-2017-0140136	12/2016	KR	N/A

OTHER PUBLICATIONS

U.S. Appl. No. 18/241,169, filed Aug. 31, 2023. cited by applicant

U.S. Appl. No. 17/976,845, filed Oct. 30, 2022, now U.S. Pat. No. 11,793,029, issued on Oct. 17, 2023. cited by applicant

U.S. Appl. No. 17/579,544, filed Jan. 19, 2022, now U.S. Pat. No. 11,515,507, issued on Nov. 29, 2022. cited by applicant

U.S. Appl. No. 17/139,476, filed Dec. 31, 2020, now U.S. Pat. No. 11,251,404, issued on Feb. 15, 2022. cited by applicant

U.S. Appl. No. 16/836,933, filed Apr. 1, 2020, now U.S. Pat. No. 10,910,600, issued on Feb. 2, 2021. cited by applicant

U.S. Appl. No. 16/241,907, filed Jan. 7, 2019, now U.S. Pat. No. 10,818,874, on Oct. 27, 2020. cited by applicant

Non-Final Office Action dated Oct. 18, 2019, issued in U.S. Appl. No. 16/241,907. cited by applicant

Notice of Allowance dated Jan. 10, 2020, issued in U.S. Appl. No. 16/241,907. cited by applicant

Notice of Allowance issued on May 13, 2020, in U.S. Appl. No. 16/241,907. cited by applicant

Final Office Action issued on Aug. 26, 2020, in U.S. Appl. No. 15/976,210. cited by applicant

Non-Final Office Action dated Jul. 9, 2020, issued in U.S. Appl. No. 16/836,933. cited by applicant

Notice of Allowance dated Sep. 30, 2020, issued in U.S. Appl. No. 16/836,933. cited by applicant

Notice of Allowance dated Oct. 8, 2020, issued in U.S. Appl. No. 17/139,476. cited by applicant

Final Office Action dated Oct. 7, 2022, issued to U.S. Appl. No. 17/406,606. cited by applicant

Notice of Allowance dated Sep. 14, 2022, issued to U.S. Appl. No. 17/579,544. cited by applicant

Non-Final Office Action dated Feb. 28, 2023, issued to U.S. Appl. No. 17/976,845. cited by applicant

Notice of Allowance dated Oct. 29, 2023, issued to Korean Patent Application No. 10-2018-0088581 (with English Concise Explanation). cited by applicant

Chinese Office Action for Chinese Patent Application or Patent No. 201910004791.8, dated Jan. 31, 2024. cited by applicant

Chinese Office Action for Chinese Patent Application or Patent No. 201810952281.9, dated Jan. 2, 2024. cited by applicant

Chinese Office Action for Chinese Patent Application No. 201910004791.8, dated Aug. 28, 2024. cited by applicant

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Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS (1) This is a continuation application of U.S. patent application Ser. No. 18/241,169 filed Aug. 31, 2023, the disclosure of which is incorporated herein by reference in its entirety. U.S. patent application Ser. No. 18/241,169 is a continuation application of U.S. patent application Ser. No. 17/976,845 filed on Oct. 30, 2022, now

U.S. Pat. No. 11,793,029 issued Oct. 17, 2023, the disclosure of which is incorporated herein by reference in its entirety. U.S. patent application Ser. No. 17/976,845 is a continuation of U.S. patent application Ser. No. 17/579,544 filed on Jan. 19, 2022, now U.S. Pat. No. 11,515,507 issued Nov. 29, 2022, the disclosure of which is incorporated herein by reference in its entirety. U.S. patent application Ser. No. 17/579,544 is a continuation of U.S. patent application Ser. No. 17/139,476 filed on Dec. 31, 2020, now U.S. Pat. No. 11,251,404, issued Feb. 15, 2022, the disclosure of which is incorporated herein by reference in its entirety. U.S. patent application Ser. No. 17/139,476 is a continuation of U.S. patent application Ser. No. 16/836,933 filed on Apr. 1, 2020, now U.S. Pat. No. 10,910,600 issued Feb. 2, 2021, the disclosure of which is incorporated herein by reference in its entirety. U.S. patent application Ser. No. 16/836,933 is a divisional of U.S. patent application Ser. No. 16/241,907 filed on Jan. 7, 2019, now U.S. Pat. No. 10,818,874 issued Oct. 27, 2020, the disclosure of which is incorporated herein by reference in its entirety. U.S. patent application Ser. No. 16/241,907 claims priority from and the benefit of Korean Patent Application No. 10-2018-0002480, filed on Jan. 8, 2018, and Korean Patent Application No. 10-2018-0088581, filed on Jul. 30, 2018, which are hereby incorporated by reference for all purposes as if fully set forth herein.

BACKGROUND

Field

(1) Exemplary embodiments of the invention relate generally to an electroluminescent device, and, more specifically, to an electroluminescent device having a light transmitting area.

Discussion of the Background

(2) Recently, because various portable types of electronic devices include a camera function, a case in which only one electronic device in which the camera function is built in is carried has increased at a rapid rate compared to a case in which a camera is separately carried.

(3) In the conventional art, because the camera is provided outside of an image display region of the electronic device, there is a tendency for a space where the electronic device may display an image to be decreased.

(4) To counter this trend, U.S. Patent Application Publication No. 2016-0337570 and others disclose a structure in which the camera is located in the display region.

(5) The above information is only for enhancement of understanding of the background of the invention and therefore it may contain information that does not form the prior art that is already known in this country to a person of ordinary skill in the art.

(6) The above information disclosed in this Background section is only for understanding of the background of the inventive concepts, and, therefore, it may contain information that does not constitute prior art.

SUMMARY

(7) When an electroluminescent device may have a light transmitting region, a problem that an internal light emitted from a pixel or an external light enters into the light transmitting region occurs. Also, when irradiating a laser for forming the light transmitting region, a problem that the pixels in a display region may be damaged occurs. In addition, when forming the light transmitting region having a through-hole structure by stacking a common intermediate layer and a common upper electrode and, then, removing portions of the common intermediate layer and the common upper electrode corresponding to a through-hole by a laser, the portions of the common intermediate layer and the common upper electrode must be clearly removed. However, there may be problems that the portions of the common intermediate layer and the common upper electrode may be not clearly removed.

(8) Exemplary embodiments are to eliminate at least one among these problems.

(9) An electroluminescent device according to exemplary embodiments may include a lower structure and an encapsulation structure. The encapsulation structure may be disposed on the lower

structure. The lower structure may include an inorganic multilayer, a planarization layer, a reflecting electrode, a pixel definition layer, an intermediate multilayer, and a semi-transparent electrode. The planarization layer may be substantially transparent and may be disposed on the inorganic multilayer. The reflecting electrode may be an individual layer disposed on the planarization layer. The pixel definition layer may be substantially transparent, may be disposed on the planarization layer and may include a pixel definition portion and a first spacer. The pixel definition portion may cover a side portion of the reflecting electrode. The first spacer may have a substantially higher height than the pixel definition portion. The reflecting electrode may not be disposed under the first spacer. The intermediate multilayer may be disposed on the reflecting electrode and may have at least one intermediate common layer. The semi-transparent electrode may be a common layer disposed on the intermediate multilayer. The lower structure may have a light transmitting region and a display region surrounding at least a portion of the light transmitting region. The first spacer and a portion of the planarization layer, the portion being disposed under the first spacer, may be included in the light transmitting region. An electroluminescent unit including the reflecting electrode, the intermediate multilayer, and the semi-transparent electrode may be included in the display region. The first spacer may not be covered by at least one selected from the group of the intermediate common layer and the semi-transparent electrode. As one example, the first spacer may not be covered by the intermediate common layer and the first spacer may not be covered by the semi-transparent electrode. As another example, the first spacer may not be covered by the intermediate common layer and the first spacer may be covered by the semi-transparent electrode. As still another example, the first spacer may be covered by the intermediate common layer and the first spacer may not be covered by the semi-transparent electrode.

(10) The pixel definition layer may further include a second spacer having substantially the same height as the first spacer and having a substantially smaller area than the first spacer. The intermediate multilayer may further include at least one intermediate individual layer. The intermediate individual layer may not cover the first and second spacers. The intermediate common layer and the semi-transparent electrode may cover the second spacer.

(11) The lower structure may further include a substrate disposed under the inorganic multilayer. The inorganic multilayer may have at least one recess disposed under the first spacer and filled with the planarization layer or at least one hole disposed under the first spacer and filled with the planarization layer.

(12) The lower structure may further include a passivation layer being a semi-conductive or conductive common layer disposed on the semi-transparent electrode. The passivation layer may not cover the first spacer.

(13) The lower structure may have a buffer region having at least a portion extending along an outline of the light transmitting region between the display region and the light transmitting region to separate the display region and the light transmitting region from each other. The lower structure may have an inorganic surface portion substantially surrounding the display region and the light transmitting region. The encapsulation structure may have an inorganic lower surface. The inorganic lower surface of the encapsulation structure may be in contact with the inorganic surface portion of the lower structure to form an inorganic-inorganic encapsulation contact region substantially surrounding the display region and the light transmitting region. For example, the inorganic-inorganic encapsulation contact region may completely surround the display region and the light transmitting region.

(14) Another electroluminescent device according to exemplary embodiments may include a lower structure and an encapsulation structure. The encapsulation structure may be disposed on the lower structure. The lower structure may include a display region, a light transmitting region, and a buffer region. The light transmitting region may have a non-through-hole structure including at least a portion surrounded by the display region. The buffer region may have at least a portion extending along an outline of the light transmitting region between the display region and the light

transmitting region to separate the display region and the light transmitting region from each other. The lower structure may further include a light blocking structure extending along the outline of the light transmitting region in the at least the portion of the buffer region.

(15) The lower structure may further include an inorganic multilayer, a planarization layer, a reflecting electrode, a pixel definition layer, an intermediate multilayer, and a semi-transparent electrode. The planarization layer may be substantially transparent and may be disposed on the inorganic multilayer. The reflecting electrode may be an individual layer disposed on the planarization layer. The pixel definition layer may be substantially transparent and may be disposed on the planarization layer to cover a side portion of the reflecting electrode. The intermediate multilayer may be disposed on the reflecting electrode. The semi-transparent electrode may be a common layer disposed on the intermediate multilayer. The pixel definition layer may have a sidewall extending along the outline of the light transmitting region in the at least the portion of the buffer region. The light blocking structure may include a semi-transparent member extending on the pixel definition layer to cover the sidewall of the pixel definition layer and being a single piece with the semi-transparent electrode.

(16) The lower structure may further include an inorganic multilayer, a planarization layer, a reflecting electrode, a pixel definition layer, an intermediate multilayer, and a semi-transparent electrode. The planarization layer may be substantially transparent and may be disposed on the inorganic multilayer. The reflecting electrode may be an individual layer disposed on the planarization layer. The pixel definition layer may be substantially transparent and may be disposed on the planarization layer to cover a side portion of the reflecting electrode. The intermediate multilayer may be disposed on the reflecting electrode. The semi-transparent electrode may be a common layer and may be disposed on the intermediate multilayer. The planarization layer and the pixel definition layer may have a sidewall extending along the outline of the light transmitting region in the at least the portion of the buffer region. The light blocking structure may have a semi-transparent member extending on the planarization layer and the pixel definition layer to cover the sidewall and being a single piece with the semi-transparent electrode.

(17) The light blocking structure may further include a reflection structure disposed on the inorganic multilayer. The sidewall may be in contact with an upper surface of the reflection structure so that the semi-transparent member may be in contact with the upper surface of the reflection structure.

(18) The intermediate multilayer may have at least one intermediate common layer. A side portion of the semi-transparent member may be substantially closer to the light transmitting region than a side portion of the intermediate common layer.

(19) The lower structure may further include an inorganic multilayer and a planarization layer. The planarization layer may be substantially transparent and may be disposed on the inorganic multilayer. The light blocking structure may include a first reflection structure disposed on the planarization layer.

(20) The lower structure may further include a reflecting electrode, a pixel definition layer, an intermediate multilayer, and a semi-transparent electrode. The reflecting electrode may be an individual layer disposed on the planarization layer. The pixel definition layer may be substantially transparent and may be disposed on the planarization layer to cover a side portion of the reflecting electrode. The intermediate multilayer may be disposed on the reflecting electrode. The semi-transparent electrode may be a common layer disposed on the intermediate multilayer. The first reflection structure may be disposed on substantially the same layer as the reflecting electrode.

(21) The planarization layer may have a first sidewall extending along the outline of the light transmitting region in the at least the portion of the buffer region.

(22) The first reflection structure may extend on the planarization layer to cover the first sidewall.

(23) The light blocking structure may further include a second reflection structure disposed on the inorganic multilayer. The first sidewall may be in contact with an upper surface of the second

reflection structure so that the first reflection structure may be in contact with the upper surface of the second reflection structure.

(24) The pixel definition layer may have a second sidewall extending along the outline of the light transmitting region in the at least the portion of the buffer region. The light blocking structure may further include a semi-transparent member extending on the pixel definition layer to cover the second sidewall and being a single piece with the semi-transparent electrode. The second sidewall may be in contact with an upper surface of the first reflection structure so that the semi-transparent member may be in contact with the upper surface of the first reflective layer.

(25) The intermediate multilayer may include at least one intermediate common layer. A side portion of the semi-transparent member may be substantially closer to the light transmitting region than a side portion of the intermediate common layer.

(26) The first reflection structure may have at least one hole.

(27) The lower structure may include an inorganic multilayer structure, a planarization layer, a reflecting electrode, a pixel definition layer, an intermediate multilayer, and a semi-transparent electrode. The planarization layer may be substantially transparent and may be disposed on the inorganic multilayer structure. The reflecting electrode may be an individual layer disposed on the planarization layer. The pixel definition layer may be substantially transparent and may be disposed on the planarization layer to cover a side portion of the reflecting electrode. The intermediate multilayer may be disposed on the reflecting electrode. The semi-transparent electrode may be a common layer disposed on the intermediate multilayer. The inorganic multilayer structure may include at least one recess corresponding to the light transmitting region and filled with the planarization layer.

(28) The light blocking structure may include a conductive material. The light blocking structure may not transmit an electrical signal.

(29) Another electroluminescent device according to exemplary embodiments may include a lower structure and an encapsulation structure. The lower structure may include a substrate and an inorganic multilayer. The substrate may include glass or an organic polymer. The inorganic multilayer may be disposed on the substrate. The encapsulation structure may be disposed on the lower structure. The lower structure may include a display region and a light transmitting region. The display region may be defined inside an outline of the inorganic multilayer in a plan view. The light transmitting region may be defined inside the outline of the inorganic multilayer in a plan view and may have a non-through-hole structure having at least a portion surrounded by the display region. An upper portion of the inorganic multilayer may have at least one recess corresponding to the light transmitting region. The display region may be substantially opaque.

(30) The inorganic multilayer may include an oxide layer and a nitride layer. The oxide layer may be partially removed by the at least one recess at the upper portion of the inorganic multilayer and the nitride layer may not be partially removed by the at least one recess at the upper portion of the inorganic multilayer.

(31) The light transmitting region may include a plurality of substantially transparent display pixels.

(32) Another electroluminescent device according to exemplary embodiments may include a lower structure and an encapsulation structure. The lower structure may include a glass substrate and an inorganic multilayer disposed on the glass substrate. The encapsulation structure may be disposed on the lower structure. The lower structure may include a display region defined inside an outline of the inorganic multilayer in a plan view and a light transmitting region defined inside an outline of the inorganic multilayer in a plan view and having a non-through-hole structure including at least a portion surrounded by the display region. The inorganic multilayer may include at least one hole corresponding to the light transmitting region. The display region may be substantially opaque.

(33) The light transmitting region may include a plurality of substantially transparent display

pixels.

(34) According to the exemplary embodiments, the light blocking structure capable of preventing the light from entering into the light transmitting region having the non-through-hole structure may be provided. By forming the light transmitting region to have the non-through-hole structure, the damage due to irradiation of laser which may be employed to form a light transmitting region having a through-hole structure may not be generated. Further, the light transmitting region may have a portion patterned like a substantial “Ω” shape.

(35) It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory and are intended to provide further explanation of the invention as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are included to provide a further understanding of the invention and are incorporated in and constitute a part of this specification, illustrate exemplary embodiments of the invention, and together with the description serve to explain the inventive concepts.

(2) FIG. 1 is a plan view illustrating an electroluminescent device according to exemplary embodiments.

(3) FIG. 2 is a cross-sectional view taken along a line II-II' in FIG. 1.

(4) FIG. 3 and FIG. 4 are plan views illustrating open masks for evaporation deposition processes according to exemplary embodiments.

(5) FIGS. 5 to 9 are enlarged views illustrating holes formed in a region B in FIG. 4.

(6) FIG. 10 and FIG. 11 are plan views illustrating electroluminescent devices according to exemplary embodiments.

(7) FIG. 12 is a plan view illustrating an individual mask for an evaporation deposition process according to exemplary embodiments.

(8) FIG. 13 and FIG. 14 are enlarged views illustrating a portion Y in FIG. 2.

(9) FIG. 15 is a cross-sectional view illustrating an electroluminescent device according to exemplary embodiments.

(10) FIG. 16 and FIG. 17 are plan views illustrating panels according to exemplary embodiments.

(11) FIG. 18 is a cross-sectional view taken along a line XVII-XVII' in FIG. 17.

(12) FIGS. 19 to 21 are plan views showing open masks for evaporation deposition processes according to exemplary embodiments.

(13) FIG. 22 and FIG. 23 are plan views illustrating an electroluminescent device formed by using an open mask shown in FIG. 19, FIG. 20, or FIG. 21 according to exemplary embodiments.

(14) FIG. 24 is a plan view illustrating an open mask for an evaporation deposition process according to exemplary embodiments.

(15) FIG. 25 is a cross-sectional view illustrating an electroluminescent device according to exemplary embodiments.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(16) In the following description, for the purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of various exemplary embodiments or implementations of the invention. As used herein “embodiments” and “implementations” are interchangeable words that are non-limiting examples of devices or methods employing one or more of the inventive concepts disclosed herein. It is apparent, however, that various exemplary embodiments may be practiced without these specific details or with one or more equivalent arrangements. In other instances, well-known structures and devices are shown in block diagram

form in order to avoid unnecessarily obscuring various exemplary embodiments. Further, various exemplary embodiments may be different, but do not have to be exclusive. For example, specific shapes, configurations, and characteristics of an exemplary embodiment may be used or implemented in another exemplary embodiment without departing from the inventive concepts.

(17) Unless otherwise specified, the illustrated exemplary embodiments are to be understood as providing exemplary features of varying detail of some ways in which the inventive concepts may be implemented in practice. Therefore, unless otherwise specified, the features, components, modules, layers, films, panels, regions, and/or aspects, etc. (hereinafter individually or collectively referred to as “elements”), of the various embodiments may be otherwise combined, separated, interchanged, and/or rearranged without departing from the inventive concepts.

(18) The use of cross-hatching and/or shading in the accompanying drawings is generally provided to clarify boundaries between adjacent elements. As such, neither the presence nor the absence of cross-hatching or shading conveys or indicates any preference or requirement for particular materials, material properties, dimensions, proportions, commonalities between illustrated elements, and/or any other characteristic, attribute, property, etc., of the elements, unless specified. Further, in the accompanying drawings, the size and relative sizes of elements may be exaggerated for clarity and/or descriptive purposes. When an exemplary embodiment may be implemented differently, a specific process order may be performed differently from the described order. For example, two consecutively described processes may be performed substantially at the same time or performed in an order opposite to the described order. Also, like reference numerals denote like elements.

(19) When an element, such as a layer, is referred to as being “on,” “connected to,” or “coupled to” another element or layer, it may be directly on, connected to, or coupled to the other element or layer or intervening elements or layers may be present. When, however, an element or layer is referred to as being “directly on,” “directly connected to,” or “directly coupled to” another element or layer, there are no intervening elements or layers present. To this end, the term “connected” may refer to physical, electrical, and/or fluid connection, with or without intervening elements. Further, the D1-axis, the D2-axis, and the D3-axis are not limited to three axes of a rectangular coordinate system, such as the x, y, and z-axes, and may be interpreted in a broader sense. For example, the D1-axis, the D2-axis, and the D3-axis may be perpendicular to one another, or may represent different directions that are not perpendicular to one another. For the purposes of this disclosure, “at least one of X, Y, and Z” and “at least one selected from the group of X, Y, and Z” may be construed as X only, Y only, Z only, or any combination of two or more of X, Y, and Z, such as, for instance, XYZ, XYY, YZ, and ZZ. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

(20) Although the terms “first,” “second,” etc. may be used herein to describe various types of elements, these elements should not be limited by these terms. These terms are used to distinguish one element from another element. Thus, a first element discussed below could be termed a second element without departing from the teachings of the disclosure.

(21) Spatially relative terms, such as “beneath,” “below,” “under,” “lower,” “above,” “upper,” “over,” “higher,” “side” (e.g., as in “sidewall”), and the like, may be used herein for descriptive purposes, and, thereby, to describe one elements relationship to another element(s) as illustrated in the drawings. Spatially relative terms are intended to encompass different orientations of an apparatus in use, operation, and/or manufacture in addition to the orientation depicted in the drawings. For example, if the apparatus in the drawings is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “above” the other elements or features. Thus, the exemplary term “below” can encompass both an orientation of above and below. Furthermore, the apparatus may be otherwise oriented (e.g., rotated 90 degrees or at other orientations), and, as such, the spatially relative descriptors used herein interpreted accordingly.

(22) The terminology used herein is for the purpose of describing particular embodiments and is

not intended to be limiting. As used herein, the singular forms, “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. Moreover, the terms “comprises,” “comprising,” “includes,” and/or “including,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, components, and/or groups thereof, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. It is also noted that, as used herein, the terms “substantially,” “about,” and other similar terms, are used as terms of approximation and not as terms of degree, and, as such, are utilized to account for inherent deviations in measured, calculated, and/or provided values that would be recognized by one of ordinary skill in the art.

(23) Various exemplary embodiments are described herein with reference to sectional and/or exploded illustrations that are schematic illustrations of idealized exemplary embodiments and/or intermediate structures. As such, variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are to be expected. Thus, exemplary embodiments disclosed herein should not necessarily be construed as limited to the particular illustrated shapes of regions, but are to include deviations in shapes that result from, for instance, manufacturing. In this manner, regions illustrated in the drawings may be schematic in nature and the shapes of these regions may not reflect actual shapes of regions of a device and, as such, are not necessarily intended to be limiting.

(24) Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure is a part. Terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and should not be interpreted in an idealized or overly formal sense, unless expressly so defined herein.

(25) An electroluminescent device according to exemplary embodiments will be described with reference to FIG. 1.

(26) FIG. 1 is a plan view illustrating an electroluminescent device according to exemplary embodiments.

(27) The electroluminescent device according to exemplary embodiments may include an external buffer region **201**, a display region **100**, a light transmitting region **210**, and an inorganic-inorganic encapsulation contact region **251**.

(28) The display region **100** may include a plurality of pixels to display an image.

(29) The light transmitting region **210** may have a light transmittance substantially higher than at least one selected from the group of the display region **100** and the external buffer region **201**. At least one selected from the group of a light incident into at least one optical unit **10** disposed under the light transmitting region **210** and a light emitted from the optical unit **10** may pass through the light transmitting region **210**. Examples of the optical unit **10** may include a camera, a flash, a sensor, etc.

(30) In exemplary embodiments, the light transmitting region **210** may have a non-hole structure. Because a size of the light transmitting region **210** may be substantially larger than a pixel circuit zone PCZ in FIG. 13, the light transmitting region **210** may be substantially different from a light transmitting zone in the pixel for realizing a transparent display.

(31) The external buffer region **201** may completely surround the display region **100** and the light transmitting region **210** in a plan view.

(32) The inorganic-inorganic encapsulation contact region **251** may completely surround the external buffer region **201** in a plan view.

(33) A portion of the external buffer region **201** may extend between the display region **100** and the light transmitting region **210** such that the display region **100** and the light transmitting region **210** may be separated from each other by the portion of the external buffer region **201**.

(34) The display region **100** may partially surround the light transmitting region **210** in a plan view.

(35) Next, a cross-sectional view taken along a line II-II' in FIG. 1 is described.

(36) FIG. 2 is a cross-sectional view taken along the line II-II' in FIG. 1.

(37) Referring to FIG. 2, the light transmitting region **210** may include a lower glass substrate **140**, a planarization layer **19**, a lower extending portion **19a** of the planarization layer **19**, a pixel definition layer **21**, a first spacer **21a** extending upward from the pixel definition layer **21**, and an upper glass substrate **150**. The lower extending portion **19a** of the planarization layer **19** may extend downward from the planarization layer **19**. The light transmitting region **210** may have no portion substantially shielding the light and, thus, the light transmitting region **210** may have a substantially high transparency. A space between the upper glass substrate **150** and the first spacer **21a** may be empty. Alternatively, the space between the upper glass substrate **150** and the first spacer **21a** may be filled with a filling structure having a substantially excellent light transmittance.

(38) A touch sensor structure **160** may be disposed on the upper glass substrate **150**. A polarization layer **39** may be disposed on the touch sensor structure **160**. A portion of the touch sensor structure **160**, the portion corresponding to the light transmitting region **210**, may be removed. A portion of the polarization layer **39**, the portion corresponding to the light transmitting region **210**, may be removed. A transparent layer may be formed on the upper glass substrate **150** to correspond to the light transmitting region **210**.

(39) The planarization layer **19** may have the lower extending portion **19a** corresponding to the light transmitting region **210**. A portion of a preliminary inorganic multilayer, the portion corresponding to the light transmitting region **210**, may be etched to form at least one hole and, then, the hole may be filled with the planarization layer **19** to form the lower extending portion **19a** of the planarization layer **19**. By etching the portion of the preliminary inorganic multilayer, the portion corresponding to the light transmitting region **210**, the preliminary inorganic multilayer may become an inorganic multilayer **26b**. According to FIG. 2, the inorganic multilayer **26b** may include a first inorganic layer **13**, a second inorganic layer **15**, and a third inorganic layer **17**. The inorganic multilayer **26b** and the lower glass substrate **140** may be collectively referred to as an inorganic multilayer structure **26a**.

(40) The pixel definition layer **21** may include the first spacer **21a**. The first spacer **21a** may correspond to the light transmitting region **210**. At least one selected from the group of an intermediate multilayer **22**, a common upper electrode **23**, and a common organic passivation layer **24** may be not formed on the first spacer **21a**. The intermediate multilayer **22** may include a common hole transport layer (hereinafter, referred to as HTL) **22a**, an individual emission layer **22b**, and a common electron transport layer (hereinafter, referred to as ETL) **22c**. According to exemplary embodiments, a layer having a high transparency among the common HTL **22a**, the common ETL **22c**, the common upper electrode **23**, and the common organic passivation layer **24** may be disposed on the first spacer **21a**. Hereinafter, at least one selected from the group of the common HTL **22a**, the common ETL **22c**, the common upper electrode **23**, and the common organic passivation layer **24** may be referred to as a common layer.

(41) The display region **100** may include a plurality of pixels, and one pixel may include a pixel circuit and a light emitting unit emitting light by using a current received from the pixel circuit. The light emitting units may be distinguished from each other by the pixel definition layer **21**, and the pixel definition layer **21** in the display region **100** may include a second spacer **21b** extending upward.

(42) An inorganic frit **250** and the upper glass substrate **150** may serve as an upper encapsulation structure preventing external moisture from flowing into the organic emission layer. A contact region between the inorganic frit **250** and the third inorganic layer **17** may correspond to the inorganic-inorganic encapsulation contact region **251**. In FIG. 2, an upper surface of the third inorganic layer **17** may have a region P in contact with the inorganic frit **250**. The region P may correspond to an inorganic surface portion P of the inorganic multilayer structure **26a** including the lower glass substrate **140** and the inorganic multilayer **26b**. The inorganic multilayer structure **26a**

may include at least one lower inorganic encapsulation layer horizontally extending, e.g., under the entire upper surface of the inorganic multilayer structure **26a** to vertically correspond to the entire upper surface of the inorganic multilayer structure **26a**. The lower glass substrate **140** may correspond to the lower inorganic encapsulation layer. Only at least one inorganic layer may be disposed between the inorganic surface portion P and the lower inorganic encapsulation layer.

(43) In exemplary embodiments, the display region **100** may be substantially opaque. Because the light transmitting region **210** may be substantially larger than a pixel circuit zone PCZ which may be occupied by the pixel circuit to drive each pixel as illustrated in FIG. **13**, the light transmitting region **210** may be substantially different from a light-transmitting zone formed in each pixel circuit zone PCZ for realizing the transparent display. As one example, the pixel circuit zone PCZ may have a substantial rectangle shape with approximate dimensions of about 25 micrometers in width and about 50 micrometers in length. As another example, the light transmitting region **210** may have a substantially circular shape having a diameter of about 3 mm. The first spacer **21a** may have a larger area than the pixel circuit zone PCZ. Alternatively, the second spacer **21b** may have a smaller area than the pixel circuit zone PCZ.

(44) In exemplary embodiments, the light transmitting region **210** may have an area substantially larger than one color unit including pixels of different colors to achieve a white light. For example, the color unit may include a red pixel, a green pixel, and a blue pixel. In the light transmitting region **210** having a shape such as a substantial “L” shape, a substantial “U” shape, or a substantial “Ω” shape shown in FIG. **1**, the area of the light transmitting region **210** may mean the area of a portion of the light transmitting region **210** substantially surrounded by the display region **100**. That the display region **100** is “substantially opaque” or “opaque” means the display region **100** has substantially insufficient light transmittance to be used as a transparent display, but does not mean that the display region **100** has a light transmittance of about zero percentage. The opaque display region **100** may include an opaque sub-display region having a first area and a transparent sub-display region having a second area that may be substantially smaller than the first area and may be completely or partially surrounded by the opaque sub-display region. Apart from the opaque display region **100**, a plurality of transparent display pixels may be formed in the light transmitting region **210**.

(45) A meaning of a through-hole structure may include a case where a hole may be formed through both the lower substrate and the lower structure. A meaning of a non-through-hole structure may include a case where a hole may not be formed through at least one selected from the group of the lower substrate and the lower structure.

(46) At least one selected from the group of the common HTL **22a**, the common ETL **22c**, the common upper electrode **23**, and the common organic passivation layer **24** may be disposed on the second spacer **21b** and may not be disposed on the first spacer **21a**.

(47) The lower electrode **20** which may be an individual layer may not overlap the light transmitting region **210**. The lower electrode **20** may include a low light transmitting material or a reflective material.

(48) A touch electrode structure **60** of the touch sensor structure **160** may not overlap the light transmitting region **210**. The touch electrode structure **60** may include the low light transmitting material.

(49) The polarization layer **39** may not overlap the light transmitting region **210**. The polarization layer **39** may have a low light transmissivity characteristic. Instead of the polarization layer **39**, a color filter CF having light-filtering regions R, G, and B and a light blocking member may be employed. Examples of the light blocking member may include a black matrix BM. When the color filter CF may be employed, the light-filtering regions R, G, and B and the black matrix BM may not overlap the light transmitting region **210**.

(50) In FIG. **2**, a touch insulation structure **65** of the touch sensor structure **160** may overlap the light transmitting region **210**. The touch insulation structure **65** may be partially removed not to

overlap the light transmitting region **210**.

(51) In FIG. **2**, the upper glass substrate **150** and the first spacer **21a** may be disposed to be spaced apart from each other. As one alternative, the upper glass substrate **150** and the first spacer **21a** may be in contact with each other. As another alternative, a filling structure having a substantially excellent light transmittance may be interposed between the upper glass substrate **150** and the first spacer **21a**.

(52) In FIG. **2**, the common organic passivation layer **24** may not cover the first spacer **21a** such that a light transmittance may be increased. The common organic passivation layer **24** may be in contact with the common upper electrode **23**. The common organic passivation layer **24** may include a semi-conductive or conductive organic material having a refractive index of over about 1.8. The common organic passivation layer **24** may serve to increase a light extraction and may serve to physically protect the common upper electrode **23**. The common organic passivation layer **24** may be formed not by a plasma-enhanced chemical vapor deposition (hereinafter, referred to as PE-CVD) method but by an evaporation deposition process so as not to damage the common upper electrode **23**.

(53) According to exemplary embodiments, the common organic passivation layer **24** may be formed to cover the first spacer **21a**. In this case, the common organic passivation layer **24** may physically protect the common upper electrode **23** more efficiently.

(54) When the common organic passivation layer **24** may be formed not to cover the first spacer **21a**, a mask in FIG. **3** and FIG. **4** may be used to form the common organic passivation layer **24**. When the common organic passivation layer **24** may be formed to cover the first spacer **21a**, an open mask without a portion A in FIG. **3** may be used to form the common organic passivation layer **24**.

(55) In FIG. **2**, the first inorganic layer **13**, the second inorganic layer **15**, and the third inorganic layer **17** which may be included in the inorganic multilayer **26b** may be all etched so that the inorganic multilayer structure **26a** may have at least one recess R.

(56) As one alternative, only the second and third inorganic layers **15** and **17** may be etched so that the inorganic multilayer structure **26a** may have the recess R. As another alternative, only the third inorganic layer **17** may be etched so that the inorganic multilayer structure **26a** may have the recess R. As still another alternative, none of the first, second, and third inorganic layers **13**, **15**, and **17** may be etched. However, if only the second and third inorganic layers **15** and **17** may be etched, only the third inorganic layer **17** may be etched, or none of the first, second, and third inorganic layers **13**, **15**, and **17** may be etched, it may be advantageous in view of light transmittance improvement that a structure lowering a light transmittance of the light transmitting region **210** may not be disposed directly on a portion of the inorganic multilayer **26b** corresponding to the light transmitting region **210** or may not be disposed inside the portion of the inorganic multilayer **26b** corresponding to the light transmitting region **210**. Examples of the structure lowering the light transmittance of the light transmitting region **210** may include a reflective wiring below the first spacer **21a**, a thin film transistor (hereinafter, referred to as TFT) below the first spacer **21a**, a capacitor below the first spacer **21a**, etc.

(57) The pixel definition portion of the pixel definition layer **21** may have a height h1, and the first spacer **21a** and the second spacer **21b** may have a height h2 that may be substantially higher than the height h1. Here, to have the height h1 may mean that an uppermost surface may be formed at the height h1.

(58) At least one selected from the group of the pixel definition layer **21** and the planarization layer **19** may include an organic material having a light transmittance of about 70% or more with respect to a wavelength of about 550 nm and a yellow color index less than about 95 at a thickness of 0.025 mm. For example, it may be preferable that the organic material may have the light transmittance of about 80% or more with respect to the wavelength of about 550 nm and the yellow color index less than about 20 at the thickness of about 0.025 mm.

(59) According to exemplary embodiments, the pixel definition layer **21** may include a colored poly-imide (hereinafter, referred to as PI) or a transparent PI, and the planarization layer **19** may include an acryl-based resin. The colored PI may include a PI having a yellowish characteristic.

(60) According to exemplary embodiments, the pixel definition layer **21** and the planarization layer **19** may include a colored PI or a transparent PI.

(61) According to exemplary embodiments, both the pixel definition layer **21** and the planarization layer **19** may include the acryl-based resin.

(62) According to exemplary embodiments, at least one selected from the group of the pixel definition layer **21** and the planarization layer **19** may include at least one selected from the group of a siloxane organic material and a silazane organic material.

(63) According to exemplary embodiments, a portion X in FIG. 2 ranging from an upper surface of the first spacer **21a** to a lower surface of the lower extending portion **19a** of the planarization layer **19** may be removed and, then, a space formed by removing the portion X in FIG. 2 may be filled with a filling structure having a substantially good light transmittance. Examples of the filling structure may include a resin. The filling structure including the resin may have substantially the same height as the first spacer **21a** such that the filling structure including the resin may perform the spacer function of the first spacer **21a**.

(64) When the inorganic multilayer **26b** exists in the light transmitting region **210**, a recess formed at a lower portion of the glass substrate **140** or a hole penetrating through the lower glass substrate **140** may be formed. An optical unit may be inserted into the recess or the hole.

(65) When the hole may be formed through the lower glass substrate **140** and the inorganic multilayer **26b** may not exist in the light transmitting region **210**, the penetration of moisture may not be prevented. Accordingly, if the inorganic multilayer **26b** does not exist in the light transmitting region **210**, only the recess, i.e., not the hole, may be formed at the lower portion of the lower glass substrate **140** such that the optical unit may be inserted in to the recess.

(66) The first inorganic layer **13** and the third inorganic layer **17** may include a silicon nitride. Examples of the silicon nitride may include SiNx, SiON, etc. The second inorganic layer **15** may include a silicon oxide. Examples of the silicon oxide may include SiO₂.

(67) Open masks according to exemplary embodiments will be described with reference to FIG. 3 and FIG. 4.

(68) First, an open mask in FIG. 3 will be described.

(69) FIG. 3 is a plan view illustrating an open mask for an evaporation deposition process to form at least one selected from the group of the common HTL **22a**, the common ETL **22c**, the common upper electrode **23**, and the common organic passivation layer **24** in FIG. 2 according to exemplary embodiments.

(70) The region A of the open mask in FIG. 3 may be adjacent to the first spacer **21a** shown in FIG. 2 in the evaporation deposition process to form at least one selected from the group of the common HTL **22a**, the common ETL **22c**, the common upper electrode **23**, and the common organic passivation layer **24** such that the at least one selected from the group of the common HTL **22a**, the common ETL **22c**, the common upper electrode **23**, and the common organic passivation layer **24** may not be formed on the first spacer **21a**.

(71) Accordingly, the light transmittance deterioration of the light transmitting region **210**, the light transmittance deterioration being generated by the at least one selected from the group of the common HTL **22a**, the common ETL **22c**, the common upper electrode **23**, and the common organic passivation layer **24** formed on the first spacer **21a**, may be prevented.

(72) An open mask in FIG. 4 will be described.

(73) FIG. 4 is a plan view illustrating an open mask used in an evaporation deposition process to form at least one selected from the group of the common HTL **22a**, the common ETL **22c**, the common upper electrode **23**, and the common organic passivation layer **24** in FIG. 2 according to exemplary embodiments.

(74) The region A of the open mask shown in FIG. 4 may be adjacent to the first spacer **21a** shown in FIG. 2 in the evaporation deposition process to form at least one selected from the group of the common HTL **22a**, the common ETL **22c**, the common upper electrode **23**, and the common organic passivation layer **24** such that the at least one selected from the group of the common HTL **22a**, the common ETL **22c**, the common upper electrode **23**, and the common organic passivation layer **24** may not be formed on the first spacer **21a**.

(75) Accordingly, the light transmittance deterioration of the light transmitting region **210**, the light transmittance deterioration being generated by the at least one selected from the group of the common HTL **22a**, the common ETL **22c**, the common upper electrode **23**, and the common organic passivation layer **24** formed on the first spacer **21a**, may be prevented.

(76) The common layer may not be formed on the second spacer **21b** overlapping the region B. The common layer may be formed on the second spacer **21b** overlapping the hole H. The common layer may be formed on the second spacer **21b** overlapping the region C. The second spacer **21b** overlapping the region B may not be formed. The second spacer **21b** overlapping the hole H may not be formed. The second spacer **21b** overlapping the region B and the second spacer **21b** overlapping the hole H may not be formed.

(77) The region B having the plurality of holes H may extend to occupy the entire region C.

(78) When the region A may be substantially circular, an ancillary hole H1 may be further formed near the region A so that the region A may have a substantially circular shape efficiently.

(79) A shape of the hole H according to exemplary embodiments will be described with reference to FIGS. 5 to 9.

(80) FIGS. 5 to 9 are enlarged views illustrating a shape of the hole H formed in the region B shown in FIG. 4.

(81) The hole H used in the open mask like FIG. 4 may have various shapes as shown in FIGS. 5 to 9.

(82) FIG. 5 shows exemplary embodiments in which the shape of each hole H may be substantially hexagonal and may have an overall honeycomb shape.

(83) In FIG. 6, the shape of each hole H may be a substantial square. In FIG. 7, the shape of each hole H may be substantially rectangular. In FIG. 6, the holes H of the substantial square shape may be arranged in rows and columns. However, in FIG. 7, the holes H of the substantially rectangular shape may be arranged in rows and may be arranged to be crossed in the columns. For example, the holes H of the substantially rectangular shape may be arranged to be offset in the column direction and may be arranged not to be offset in the row direction.

(84) In exemplary embodiments illustrated in FIG. 8, the shape of each hole H may be triangular shape, while in exemplary embodiments illustrated in FIG. 9, the shape of each hole H may be substantially circular, and an ancillary hole H2 may be further formed.

(85) As shown in the exemplary embodiments illustrated in FIGS. 5 to 8, the holes may have various shapes, such as hexagonal, quadrangular, triangular, etc., allowing the intervals between the holes H to be substantially constant.

(86) When the shape of the hole may be substantially circular like in FIG. 9, in the space between the substantially circular shapes, a substantially non-circular ancillary holes H2 having a shape such as a triangle may be further formed.

(87) The hole H may be formed to vertically correspond to the light emission zone of each pixel.

(88) A size of the hole H may be sufficient enough to allow at least one selected from the group of the common HTL **22a**, the common ETL **22c**, and the common upper electrode **23** to become the common layer by using a shadow effect. The shadow effect may mean a phenomenon generated during the evaporation deposition of the organic material through a shadow mask having a hole. When the shadow effect may be generated, a deposited region of the organic material deposited through the hole may be substantially larger than an area of the hole and an outline of the hole may be disposed inside an outline the deposited region.

(89) When determining a size of the hole H for forming at least one selected from the group of the common HTL **22a** and the common ETL **22c**, the hole H may have a size substantially smaller than a predetermined size allowing the at least one selected from the group of the common HTL **22a** and the common ETL **22c** to become a common layer. Even when the at least one of the common HTL **22a** and the common ETL **22c** may become a partially common layer or an individual layer, the electroluminescent device may operate normally. However, when determining a size of the hole H for forming the common upper electrode **23**, the hole H may have a size no less than a predetermined size allowing the common upper electrode **23** to become a common layer. When the common upper electrode **23** may become a partially common layer or an individual layer, the electroluminescent device may not operate normally.

(90) An electroluminescent device according to exemplary embodiments will be described with reference to FIG. **10** and FIG. **11**.

(91) FIG. **10** and FIG. **11** are plan views illustrating electroluminescent devices each having a display region, a first spacer, a third spacer, a fourth spacer, and an inorganic-inorganic encapsulation contact region according to exemplary embodiments.

(92) Referring to FIG. **10**, a third spacer **21c** may be connected between the first spacer **21a** and a fourth spacer **21d**. The first spacer **21a** may be formed to correspond to the light transmitting region **210**. The fourth spacer **21d** may be disposed inside the inorganic-inorganic encapsulation contact region **251** and may be formed along the outline of the display region **100**. The fourth spacer **21d** may surround the display region **100** in a plan view. The fourth spacer **21d** may be surrounded by the inorganic-inorganic encapsulation contact region **251** in a plan view. The third spacer **21c** may extend from the fourth spacer **21d** toward the light transmitting region **210** and may be connected with the first spacer **21a**.

(93) The third spacer **21c** and the fourth spacer **21d** may have substantially the same height as the first spacer **21a**. The first spacer **21a**, the third spacer **21c**, and the fourth spacer **21d** may be integrally formed as a single piece.

(94) When comparing FIG. **10** to FIG. **3**, the region A in FIG. **3** may correspond to the first spacer **21a** such that the region A in FIG. **3** may be in contact with or may be adjacent to the first spacer **21a**, the region P may correspond to the fourth spacer **21d** such that the region P may be in contact with or may be adjacent to the fourth spacer **21d**, and the region disposed between the region A and the region P may correspond to the third spacer **21c** such that the region disposed between the region A and the region P may be in contact with or may be adjacent to the third spacer **21c**.

(95) Referring to FIG. **11**, the first spacer **21a** and the third spacer **21c** may be connected. The inorganic-inorganic encapsulation contact region **251** may surround the display region **100** while passing between the third spacer **21c** and the fourth spacer **21d** in a plan view. The third spacer **21c** and the fourth spacer **21d** may have substantially the same height as the first spacer **21a**. The first spacer **21a** and the third spacer **21c** may be integrally formed as a single piece. The fourth spacer **21d** may have a shape such that the fourth spacer **21d** may surround the inorganic-inorganic junction encapsulation region **251** in a plan view.

(96) When comparing FIG. **11** to FIG. **3**, the region A in FIG. **3** may correspond to the first spacer **21a** such that the region A in FIG. **3** may be in contact with or may be adjacent to the first spacer **21a**, the region P may correspond to the fourth spacer **21d**, such that the region P may be in contact with or may be adjacent to the fourth spacer **21d**, and the region disposed between the region A and the region P may correspond to the third spacer **21c** such that the region disposed between the region A and the region P may be in contact with or may be adjacent to the third spacer **21c**.

(97) An individual mask according to exemplary embodiments will be described with reference to FIG. **12**.

(98) FIG. **12** is a plan view illustrating an individual mask for an evaporation deposition process to form an individual emission layer in FIG. **2** according to exemplary embodiments.

(99) Referring to FIG. **12**, a region A of the individual mask may be adjacent to the first spacer **21a**

in FIG. 2 during the evaporation deposition process for forming the individual emission layer **22b** such that the individual emission layer **22b** may not be formed on the first spacer **21a**.

(100) Accordingly, a light transmittance deterioration of the light transmitting region **210**, the light transmittance deterioration being generated by the individual emission layer **22b** formed on the first spacer **21a**, may be prevented. When the individual emission layer **22b** may have a color and may include a transition metal, the individual emission layer **22b** may exceedingly deteriorate a light transmittance. Therefore, it may be preferable that the individual emission layer **22b** having the color and including the transition metal may not be formed in the light transmitting region **210**.

(101) The region B of the individual mask during the evaporation deposition process for forming the individual emission layer **22b** may be disposed to be adjacent to the second spacer **21b** in FIG. 2. Accordingly, the individual emission layer **22b** may not be formed on the second spacer **21b**.

(102) A method for solving a light interference problem due to a light advancing from the individual emission layer **22b** to the optical unit will be described with reference to FIG. 13.

(103) FIG. 13 is an enlarged view illustrating a portion Y in FIG. 2 according to exemplary embodiments.

(104) In FIG. 13, a pixel circuit zone PCZ may include a TFT. The TFT may include a semiconductor layer **12**, a gate electrode **14**, a source electrode **18s**, and a drain electrode **18d**. The semiconductor layer **12** may be disposed on the first inorganic layer **13**. The semiconductor layer **12** may include a channel region **12c**, a source region **12s**, and a drain region **12d**. The source and drain regions **12s** and **12d** may be disposed on opposite sides of the channel region **12c**. The semiconductor layer **12** may be covered by the second inorganic layer **15**. The gate electrode **14** may be disposed on the second inorganic layer **15** such that the gate electrode **14** may vertically correspond to the channel region **12c** of the semiconductor layer **12**. The gate electrode **14** may be covered by the third inorganic layer **17**. The third inorganic layer **17** and the second inorganic layer **15** may have openings overlapping the source and drain regions **12s** and **12d** of the semiconductor layer **12**. The source region **12s** and the drain region **12d** may be electrically connected to the source electrode **18s** and the drain electrode **18d**, respectively, through the openings. One pixel may include a plurality of TFTs, and one TFT among them may be electrically connected to a lower electrode **20**. The lower electrode **20** may be connected to the drain electrode **18d** of the TFT. The organic emission layer may be disposed on the lower electrode **20**. The organic emission layer may include the common HTL **22a**, the individual emission layer **22b**, and the common ETL **22c**. The common upper electrode **23** may be disposed on the organic emission layer. The individual emission layer **22b** in the organic emission layer may emit a light for displaying an image by a current flowing between the lower electrode **20** and the common upper electrode **23**.

(105) A first reflection structure **18r** may be formed by substantially the same process as the source and drain electrodes **18s** and **18d**. The first reflection structure **18r** may be formed on substantially the same layer as the source and drain electrodes **18s** and **18d**. The source and drain electrodes **18s** and **18d** may have a triple layered structure including a reflective material. The triple layered structure may include a titanium (Ti) layer, an aluminum (Al) layer on the Ti layer, and a Ti layer on the Al layer.

(106) A second reflection structure **20r** may be formed by substantially the same process as the lower electrode **20**. The second reflection structure **20r** may be formed on substantially the same layer as the lower electrode **20**. The lower electrode **20** may have a triple layered structure including a reflective material. The triple layered structure may include an indium tin oxide (ITO) layer, a silver (Ag) layer on the ITO layer, and an ITO layer on the Ag layer.

(107) The first reflection structure **18r** and the second reflection structure **20r** may be in contact with each other. The first reflection structure **18r** and the second reflection structure **20r** may form the light blocking structure **170**. An external light from the outside and an inner light from the individual emission layer **22b** may advance into the optical unit **10** through the planarization layer **19**. The light blocking structure **170** may block the external light and the inner light.

- (108) The second reflection structure **20r** may have a plurality of holes. The holes may become a path for outgassing when the planarization layer **19** may be formed by using an organic material.
- (109) The first reflection structure **18r** may be omitted according to exemplary embodiments. In this case, the second reflection structure **20r** may be connected to the inorganic multilayer **26b**.
- (110) The common upper electrode **23** may include magnesium (Mg) and silver (Ag). The common upper electrode **23** may have a sufficiently thin thickness such that the common upper electrode **23** may have a semi-transparent or transfective characteristic. Because the common upper electrode **23** may have the semi-transparent or transfective characteristic, the common upper electrode **23** may have a substantially lesser light blocking effect than a single transparent electrode. However, because the light blocking effect may be substantially smaller than the reflecting electrode, at least one first groove **24a** may be formed in the pixel definition layer **21** over the planarization layer **19** or at least one second groove **24b** may be formed on the second reflection structure **20r**, thereby more effectively preventing light from flowing to the optical unit through the pixel definition layer **21**.
- (111) The common upper electrode **23** may be formed on a left sidewall, a bottom and a right sidewall of the first groove **24a**. The light passing through the first groove **24a** from a left side of the first groove **24a** to a right side of the first groove **24a** may be reduced two times by a portion of the common upper electrode **23** on the left sidewall of the first groove **24a** and a portion of the common upper electrode **23** on the right sidewall of the first groove **24a**. Therefore, the light passing through the first groove **24a** from the left side of the first groove **24a** to the right side of the first groove **24a** may be effectively blocked even though the common upper electrode **23** may have the semi-transparent or transfective characteristic. Similarly, the common upper electrode **23** may be formed on a left sidewall, a bottom and a right sidewall of the second groove **24b**. The light passing through the second groove **24b** from a left side of the second groove **24b** to a right side of the second groove **24b** may be reduced two times by a portion of the common upper electrode **23** on the left sidewall of the second groove **24b** and a portion of the common upper electrode **23** on the right sidewall of the second groove **24b**. Therefore, the light passing through the second groove **24b** from the left side of the second groove **24b** to the right side of the second groove **24b** may be effectively blocked even though the common upper electrode **23** may have the semi-transparent or transfective characteristic. Furthermore, when the pixel definition layer **21** may have the first and second grooves **24a** and **24b**, the light may be more efficiently blocked.
- (112) The first reflection structure **18r**, the second reflection structure **20r**, the first groove **24a**, and the second groove **24b** may be formed to extend along the circumference of the light transmitting region **210** between the display region **100** and the light transmitting region **210** in a plan view, thereby preventing the light being transmitted to the light transmitting region **210**.
- (113) The common upper electrode **23** may be in contact with the second reflection structure **20r** such that the second reflection structure **20r** may perform a function of ancillary wiring supplying power to the common upper electrode **23**. Alternatively, the second reflection structure **20r** may be in an electrically floating state.
- (114) In order for the common upper electrode **23** and the second reflection structure **20r** to contact each other, the region A shown in FIG. 3 of the mask for forming the common upper electrode **23** may be substantially smaller than the region A shown in FIG. 3 of the mask for forming the common ETL **22c** and the common HTL **22a**.
- (115) In FIG. 13, the common organic passivation layer **24** may be omitted. However, the common organic passivation layer **24** may be formed on the common upper electrode **23**. Also, according to exemplary embodiments, the common inorganic passivation layer (not shown) may be further formed on the common organic passivation layer **24**.
- (116) At least one light blocking structure **170** may be included in the panel to reinforce the light blocking.
- (117) In another exemplary embodiment, to reduce the light interference of the light from the

individual emission layer **22b** into the optical unit **10**, the distance between the individual emission layer **22b** and the light transmitting region **210** may increase. In this case, the light inflow to the optical unit may be reduced.

(118) In another exemplary embodiment, the light inflow may be reduced by forming the pixel near the first spacer **21a** in a normally black mode, or by allowing the pixel near the first spacer **21a** to display the black when the optical unit **10** may be used.

(119) In FIG. **13**, the position of the first groove **24a** may be substantially farther away from the light transmitting region **210** than the second groove **24b**. However, unlike in FIG. **13**, the positions of the first groove **24a** and the second groove **24b** may be switched. One first groove **24a** may be formed inner than the second groove **24b** and another first groove **24a** may be formed outer than the second groove **24b**.

(120) A variation of the exemplary embodiment in FIG. **13** will be described with reference to FIG. **14**.

(121) FIG. **14** is an enlarged view illustrating the portion Y in FIG. **2** according to exemplary embodiments. The exemplary embodiments in FIG. **14** and the exemplary embodiments in FIG. **13** may be mutually cooperative and complement each other. The exemplary embodiment in FIG. **13** and the exemplary embodiments in FIG. **14** may be not mutually exclusive.

(122) The first groove **24a** may extend to the planarization layer **19** to block the light transmitted to the optical unit **10** through the planarization layer **19**. A void may be formed in the first groove **24a** so that an effect of blocking the light by using the common upper electrode **23** may be obtained at least two times.

(123) Although not shown in FIG. **14**, the second reflection structure **20r** may be exposed from the bottom of the first groove **24a**, and the second reflection structure **20r** may be connected to the common upper electrode **23**. The power transmitted to the second reflection structure **20r** may be provided to the common upper electrode **23**, thereby the second reflection structure **20r** may also perform a function of an auxiliary electrode reducing a voltage drop, i.e., IR drop, phenomenon of the common upper electrode **23**.

(124) A void V may be formed inside the second reflection structure **20r**. The void V may be filled with the pixel definition layer **21**. The light may be blocked at least two times by the second reflection structure **20r** having the void V filled with the pixel definition layer **21** such that the light may not be transmitted to the light transmitting region **210**.

(125) Like the second reflection structure **20r** shown in FIG. **13**, the void V may also be formed inside the second reflection structure **20r** shown in FIG. **14**. In this case, the second reflection structure **20r**, the common upper electrode **23**, the common upper electrode **23**, and the second reflection structure **20r** may effectively perform light blocking functions, respectively, such that the light in the planarization layer **19** may not flow from the planarization layer **19** to the optical unit **10**.

(126) An electroluminescent device according to exemplary embodiments will be described with reference to FIG. **15**.

(127) FIG. **15** is a cross-sectional view illustrating an electroluminescent device according to exemplary embodiments.

(128) In FIG. **15**, a multi-layered encapsulation layer **31** may be employed instead of the inorganic frit **250** and the upper glass substrate **150** in FIG. **2**. A lower transparent organic polymer substrate **140a** may be employed instead of the lower glass substrate **140**. Unlike the upper glass substrate **150** and the lower glass substrate **140** in FIG. **2**, the lower transparent organic polymer substrate **140a** in FIG. **15** may include a plastic material having a flexible characteristic such as a polyimide (PI). Here, the multi-layered encapsulation layer **31** may include first and second encapsulation inorganic layers **31a** and **31c**, and may include one organic layer **31b**. The one organic layer **31b** may be disposed between the first encapsulation inorganic layer **31a** and the second inorganic encapsulation layer **31c**.

(129) In the manufacturing process, a sacrificial glass substrate (not shown) may be disposed under the lower transparent organic polymer substrate **140a**, and the sacrificial glass substrate may be removed in the final step such that the lower transparent organic polymer substrate **140a** may be located at the bottom.

(130) The light transmitting region **210** according to FIG. **15** may include the lower transparent organic polymer substrate **140a**, the lower extending portion **19a** of the planarization layer **19**, the planarization layer **19**, the pixel definition layer **21**, the first spacer **21a**, and the multi-layered encapsulation layer **31**. Therefore, the light transmitting region **210** may have the characteristic of an increased transparency.

(131) In FIG. **15**, a region where an inorganic upper surface of the third inorganic layer **17** disposed at the uppermost portion of the inorganic multilayer **26b** and an inorganic lower surface of the first encapsulation inorganic layer **31a** disposed at the lowest portion of the multi-layered encapsulation layer **31** may be in contact with each other may correspond to the inorganic-inorganic encapsulation contact region **251** which may have a function substantially the same as the inorganic frit **250** shown in FIG. **2**. The inorganic upper surface of the third inorganic layer **17** may be disposed not to be substantially lower than a plane on which a lower surface of the lower extending portion **19a** of planarization layer **19** may be disposed.

(132) In FIG. **15**, if the first inorganic layer **13**, the second inorganic layer **15**, and the third inorganic layer **17** may be all etched in the light transmitting region **210**, the lower transparent organic polymer substrate **140a** which may include an organic material not an inorganic material may be exposed. Accordingly, the moisture and oxygen may be supplied directly to an electroluminescent unit by passing through the lower transparent organic polymer substrate **140a**. Therefore, it may be preferred that the first inorganic layer **13**, the second inorganic layer **15**, and the third inorganic layer **17** may not all be etched in the light transmitting region **210**. As one example, only the third inorganic layer **17** may be etched in the light transmitting region **210**. As another example, only the second inorganic layer **15** and the third inorganic layer **17** may be etched in the light transmitting region **210**. As still another example, none of the first inorganic layer **13**, the second inorganic layer **15**, and the third inorganic layer **17** may be etched in the light transmitting region **210**.

(133) According to exemplary embodiments, when at least one recess R may be formed in the inorganic multilayer **26b** including the first inorganic layer **13**, the second inorganic layer **15**, and the third inorganic layer **17**, a thickness T2 of a portion of the inorganic multilayer **26b** where the recess R may not be formed may be substantially greater than a thickness T1 of a portion of the inorganic multilayer **26b** where the recess R may be formed.

(134) According to exemplary embodiments, a space into which the optical unit **10** may be inserted may be formed by forming a recess at a lower portion of the lower transparent organic polymer substrate **140a** or by forming a hole through the lower transparent organic polymer substrate **140a**.

(135) The first inorganic layer **13** and the third inorganic layer **17** may include a silicon nitride. The second inorganic layer **15** may include a silicon oxide. In this case, a portion of the inorganic multilayer **26b** where the recess R may be formed may have a remaining part RP only including an inorganic layer of a silicon nitride. When the portion of the inorganic multilayer **26b** where the recess R may be formed may have the remaining part RP only having an inorganic layer of a silicon oxide, the inorganic multilayer **26b** may not efficiently prevent a water penetration. This may be because the silicon oxide may have a weaker function of water penetration prevention than the silicon nitride. Particularly, as in FIG. **15**, when the lower transparent organic polymer substrate **140a** having a substantially poor water penetration prevention characteristic may be located under the inorganic multilayer **26b**, it may be required that the portion of the inorganic multilayer **26b** where the recess R may be formed may have the remaining part RP having the inorganic layer of the silicon nitride which may have a substantially good water penetration prevention characteristic.

(136) Although not shown in FIG. **15**, in order to form the recess R, the second inorganic layer **15**

including the silicon oxide may be removed and the first and third inorganic layer **13** and **17** including the silicon nitride may not be removed. In this case, the first inorganic layer **13** and the third inorganic layer **17** may be in direct contact with each other near the recess R such that the encapsulation function may be improved and the Bragg mirror phenomenon generated when several layers having substantially different refractive indexes may be deposited may be advantageously reduced.

(137) An electroluminescent device including a light transmitting region **210** having a different shape will be described with reference to FIG. **16**.

(138) FIG. **16** is a plan view illustrating an electroluminescent device according to exemplary embodiments.

(139) The electroluminescent device shown in FIG. **16** may be substantially the same as the electroluminescent device shown in FIG. **1**, except for the light transmitting region **210** having a sufficient size such that a plurality of the optical units **10** shown in FIG. **2** may be accommodated.

(140) At least two optical units **10** may be disposed on the rear surface in the light transmitting region **210** in FIG. **16**. The electroluminescent device may further include a light blocking structure (not shown) disposed in a region between the two optical units **10**. In this case, the light blocking structure may have substantially the same structure as the light blocking structure **170** shown in FIG. **13** or FIG. **14**.

(141) An electroluminescent device including a light transmitting region **210** having a different position will be described with reference to FIG. **17**.

(142) FIG. **17** is a plan view illustrating an electroluminescent device according to exemplary embodiments.

(143) Referring to FIG. **17**, an inner buffer region **202** may be further included. The inner buffer region **202** may completely surround the light transmitting region **210** in a plan view. The inner buffer region **202** may be completely surrounded by the display region **100** in a plan view. The external buffer region **201** may completely surround the display region **100**. The inorganic-inorganic encapsulation contact region **251** may completely surround the external buffer region **201** in a plan view.

(144) Exemplary embodiments will be described in detail with reference to a cross-sectional view taken along a line XVII-XVII' in FIG. **17**.

(145) FIG. **18** is a cross-sectional view taken along the line XVII-XVII' in FIG. **17**.

(146) When explaining FIG. **18**, the description of substantially the same parts as described in FIG. **2** will be omitted.

(147) At least one hole corresponding to the light transmitting region **210** may be formed through the inorganic multilayer **26b**. Because the lower glass substrate **140** may include an inorganic material such as a glass, the lower glass substrate **140** may have an excellent encapsulation characteristic unlike the lower transparent organic polymer substrate **140a** in FIG. **15** which may include the organic material. Therefore, the hole corresponding to the light transmitting region **210** may be formed through the inorganic multilayer **26b**.

(148) In exemplary embodiments, the first inorganic layer **13** may not be removed and the second and third inorganic layers **15** and **17** may be removed in the light transmitting region **210** such that the inorganic multilayer **26b** may have at least one recess corresponding to the light transmitting region **210**. In detail, when the inorganic multilayer **26b** may have a first thickness in the light transmitting region **210**, the inorganic multilayer **26b** may have a second thickness substantially thicker than the first thickness in a region except for the light transmitting region **210**.

(149) An open mask used in the exemplary embodiment in FIG. **18** will be described with reference to FIGS. **19** to **21**.

(150) FIG. **19** is a plan view illustrating an open mask for an evaporation deposition process used to form at least one selected from the group of the common HTL **22a**, the common ETL **22c**, the common upper electrode **23**, and the common organic passivation layer **24** in FIG. **18** according to

exemplary embodiments.

(151) In the evaporation deposition process for forming the at least one selected from the group of the common HTL **22a**, the common ETL **22c**, the common upper electrode **23**, and the common organic passivation layer **24**, a region A of the open mask may be adjacent to the first spacer **21a** in FIG. **18** such that the at least one selected from the group of the common HTL **22a**, the common ETL **22c**, the common upper electrode **23**, and the common organic passivation layer **24** may not be formed on the first spacer **21a**. Accordingly, a light transmittance deterioration of the light transmitting region **210**, the light transmittance deterioration being generated when the at least one selected from the group of the common HTL **22a**, the common ETL **22c**, the common upper electrode **23**, and the common organic passivation layer **24e** may be formed on the first spacer **21a**, may be prevented.

(152) In exemplary embodiments in FIG. **19**, a connecting bar **180** may be provided. A plurality of holes H may be formed in the connecting bar **180**. A deposition material entering through the holes H may be somewhat spread by the shadow effect on a portion covered by the connecting bar **180** such that the at least one selected from the group of the common HTL **22a**, the common ETL **22c**, the common upper electrode **23**, and the common organic passivation layer **24** may be commonly deposited.

(153) The at least one selected from the group of the common HTL **22a**, the common ETL **22c**, the common upper electrode **23**, and the common organic passivation layer **24** may not be formed on the second spacer **21b** overlapping the connecting bar **180**. The at least one selected from the group of the common HTL **22a**, the common ETL **22c**, the common upper electrode **23**, and the common organic passivation layer **24** may be formed on the second spacer **21b** overlapping the holes H. The at least one selected from the group of the common HTL **22a**, the common ETL **22c**, the common upper electrode **23**, and the common organic passivation layer **24** may be formed on the second spacer **21b** overlapping the region C.

(154) In FIG. **19**, the connecting bar **180** may be connected to a region A in an X-axis direction. As one alternative, the connecting bar **180** may be connected to the region A in a Y-axis direction substantially perpendicular to the X-axis direction. As another alternative, the connecting bar **180** may be connected to the region A in the X-axis direction and the connecting bar **180** may be also connected to the region A in the Y-axis direction.

(155) An open mask according to exemplary embodiments will be described with reference to FIG. **20**.

(156) FIG. **20** is a plan view illustrating an open mask for an evaporation deposition process to form at least one selected from the group of the common HTL **22a**, the common ETL **22c**, the common upper electrode **23**, and the common organic passivation layer **24** according to exemplary embodiments.

(157) The open mask in FIG. **20** may be substantially the same as the open mask described in FIG. **19** except for a single connecting bar **180**. The single connecting bar **180** may be disposed between a region P and a region A.

(158) A width of the connecting bar **180** shown in FIG. **20** may be substantially larger than a diameter of the region A. As one alternative, the width of the connecting bar **180** may be substantially equal to the diameter of the region A. As another alternative, the width of the connecting bar **180** may be substantially smaller than the diameter of the region A.

(159) It may be preferable that the connecting bar **180** may be connected to the peripheral region closest to the region A.

(160) According to exemplary embodiments, the holes H may not be formed in the region D in FIG. **20**. In this case, the display region **100** having a portion patterned like a substantial “Ω” shown in FIG. **1** may be formed.

(161) An open mask according to exemplary embodiments will be described with reference to FIG. **21**.

(162) FIG. 21 is a plan view illustrating an open mask for an evaporation deposition process to form at least one selected from the group of the common HTL 22a, the common ETL 22c, the common upper electrode 23, and the common organic passivation layer 24 in FIG. 18 according to exemplary embodiments.

(163) In the evaporation deposition process for forming at least one selected from the group of the common HTL 22a, the common ETL 22c, the common upper electrode 23, and the common organic passivation layer 24, a region A of the open mask may be adjacent to the first spacer 21a in FIG. 18 such that the at least one selected from the group of the common HTL 22a, the common ETL 22c, the common upper electrode 23, and the common organic passivation layer 24 may not be formed on the first spacer 21a. Accordingly, the light transmittance deterioration of the light transmitting region 210, the light transmittance deterioration being generated when the at least one selected from the group of the common HTL 22a, the common ETL 22c, the common upper electrode 23, and the common organic passivation layer 24e may be formed on the first spacer 21a, may be prevented.

(164) A region B, i.e., a rigid portion, having a plurality of holes H may be formed in a peripheral region of a region A such that at least one selected from the group of the common HTL 22a, the common ETL 22c, the common upper electrode 23, and the common organic passivation layer 24 may be deposited through the holes H.

(165) The at least one selected from the group of the common HTL 22a, the common ETL 22c, the common upper electrode 23, and the common organic passivation layer 24 may not be formed on the second spacer 21b overlapping the region B. The at least one selected from the group of the common HTL 22a, the common ETL 22c, the common upper electrode 23, and the common organic passivation layer 24 may be formed on the second spacer 21b overlapping the hole H.

(166) An electroluminescent device formed by using the mask in FIGS. 19 to 21 will be described with reference to FIG. 22 and FIG. 23.

(167) FIG. 22 and FIG. 23 are plan views illustrating an electroluminescent device having a display region 100, a first spacer 21a, a fourth spacer 21d, and an inorganic-inorganic encapsulation contact region 251 and formed by using the open mask shown in FIG. 19, FIG. 20, or FIG. 21 according to exemplary embodiments.

(168) Referring to FIG. 22, the first spacer 21a may be surrounded by the display region 100 in a plan view. The inorganic-inorganic encapsulation contact region 251 may surround the display region 100 in a plan view. The fourth spacer 21d may surround the inorganic-inorganic encapsulation contact region 251 in a plan view. The first spacer 21a and the fourth spacer 21d may have substantially the same height. The region A in FIGS. 19 to 21 may correspond to the first spacer 21a such that the region A may be in contact with or may be adjacent to the first spacer 21a. The region P in FIGS. 19 to 21 may correspond to the fourth spacer 21d such that the region P may be in contact with or may be adjacent to the fourth spacer 21d.

(169) Referring to FIG. 23, the first spacer 21a may be surrounded by the display region 100 in a plan view. The display region 100 may be surrounded by the fourth spacer 21d in a plan view. The fourth spacer 21d may have substantially the same height as the first spacer 21a. The fourth spacer 21d may be surrounded by the inorganic-inorganic encapsulation contact region 251 in a plan view. The fourth spacer 21d may surround the inorganic-inorganic encapsulation contact region 251 in a plan view. The first spacer 21a and the fourth spacer 21d may have substantially the same height. The region A in FIGS. 19 to 21 may correspond to the first spacer 21a such that the region A may be in contact with or adjacent to the first spacer 21a. The region P in FIGS. 19 to 21 may correspond to the fourth spacer 21d such that the region P may be in contact with or may be adjacent to the fourth spacer 21d.

(170) Hereinafter, an open mask in FIG. 18 according to exemplary embodiments will be described with reference to FIG. 24.

(171) FIG. 24 is a plan view illustrating an open mask for an evaporation deposition process to

form at least one selected from the group of the common HTL **22a**, the common ETL **22c**, the common upper electrode **23**, and the common organic passivation layer **24** in FIG. **18** according to exemplary embodiments.

(172) When using the open mask in FIG. **24**, at least one selected from the group of the common HTL **22a**, the common ETL **22c**, the common upper electrode **23**, and the common organic passivation layer **24** may be disposed on the spacer **21a** in FIG. **18**. However, when considering that the common HTL **22a**, the common ETL **22c**, and the common organic passivation layer **24** may include organic materials and the common upper electrode **23** may be a semi-transparent electrode for a top emission type, a light transmittance of the light transmitting region **210** may not be excessively reduced. Because an individual mask substantially similar to the individual mask in FIG. **12** except for a position of the region A may be used here, the individual emission layer **22b** may not be formed on the first spacer **21a**. Therefore, the open mask like FIG. **24** may be used.

(173) Exemplary variations of FIG. **18** will be described with reference to FIG. **25**.

(174) FIG. **25** is a cross-sectional view illustrating an electroluminescent device according to exemplary embodiments.

(175) Referring to FIG. **25**, a multi-layered encapsulation layer **31** may be employed instead of the inorganic frit **250** and the upper glass substrate **150** in FIG. **18**. The lower transparent organic polymer substrate **140a** may be employed instead of the lower glass substrate **140** in FIG. **18**. FIG. **25** may be substantially the same as FIG. **18** except for the above-described aspects and, thus, the description of the same parts will be omitted.

(176) In the exemplary embodiment in FIG. **25**, a second thickness T2 of a second portion of the inorganic multilayer **26b**, the second portion including the first, second and third inorganic layers **13**, **15**, and **17**, may be substantially greater than a first thickness T1 of a first portion of the inorganic multilayer **26b**, the first portion having a recess.

(177) In FIG. **25**, an inorganic passivation layer may not be shown on the common organic passivation layer **24**. However, the inorganic passivation layer may be formed on the common organic passivation layer **24**. The inorganic passivation layer may include a semi-conductive material or a conductive material. Examples of a material included in the inorganic passivation layer may include a lithium fluoride (LiF). The LiF may be a material capable of being deposited by an evaporation deposition process. Therefore, when the LiF may be deposited by the evaporation deposition process, the work function of the common upper electrode **23** may not be changed. This may be because plasma damaging to the common upper electrode **23** in the PE-CVD process may not occur in the evaporation process. The inorganic passivation layer may be in contact with the common organic passivation layer **24**. The inorganic passivation layer may be a common layer covering the first spacer **21a** to strongly protect the common upper electrode **23**. Alternatively, the inorganic passivation layer may be a common layer not covering the first spacer **21a** to improve the light transmittance of the light transmitting region **210**. When the inorganic passivation layer may be the common layer covering the first spacer **21a**, the mask shown in FIG. **24** may be used to form the inorganic passivation layer. When the inorganic passivation layer may be the common layer not covering the first spacer **21a**, the mask described in FIGS. **19** to **21** may be used to form the inorganic passivation layer.

(178) According to exemplary embodiments, the electroluminescent device may have at least two light transmitting regions **210** in FIG. **1**, at least two light transmitting regions **210** in FIG. **16**, or at least two light transmitting regions **210** in FIG. **17**. Alternatively, the electroluminescent device may have at least two different kinds of light transmitting regions **210** among the light transmitting region **210** in FIG. **1**, the light transmitting region **210** in FIG. **16**, and the light transmitting region **210** in FIG. **17**.

(179) Although certain exemplary embodiments and implementations have been described herein, other embodiments and modifications will be apparent from this description. Accordingly, the inventive concepts are not limited to such embodiments, but rather to the broader scope of the

appended claims and various obvious modifications and equivalent arrangements as would be apparent to a person of a person of ordinary skill in the art.

Claims

1. A display device, comprising: a substrate comprising a glass or a plastic; a lower structure located on the substrate, the lower structure comprising: an inorganic multilayer located on the substrate; a planarization layer located on the inorganic multilayer; first electrodes located on the planarization layer; a definition layer located on the planarization layer, the definition layer having a covering portion covering a lateral portion of the first electrode to define an exposed portion of the first electrode; emission layers located on the exposed portions of the first electrodes; and a second electrode located on the emission layers; and an upper encapsulator located on the lower structure, wherein the lower structure has a first region and a second region surrounding the first region in plan view, the first region having a light transmittance greater than the second region, the first region having a non-through-hole structure in such a way that a stack comprising the substrate and the lower structure does not have a hole corresponding to the first region, the upper encapsulator overlaps the first region in plan view, the first region comprises a plurality of first display units, the lower structure has an inorganic surface portion, the upper encapsulator having an inorganic lower surface contacting the inorganic surface portion of the lower structure, and the second region comprises a plurality of second display units.
2. The display device of claim 1, wherein the substrate comprises the glass, the inorganic multilayer is located directly on the substrate, and the inorganic multilayer comprises at least one hole corresponding to the first region.
3. The display device of claim 2, wherein the at least one hole of the inorganic multilayer is at least partially filled with a filling structure comprising a transparent resin, and a top of the filling structure is higher than a top of the planarization layer.
4. The display device of claim 1, wherein an upper portion of the inorganic multilayer has at least one recess corresponding to the first region.
5. The display device of claim 4, wherein the at least one recess of the inorganic multilayer is at least partially filled with a filling structure comprising a transparent resin, and a top of the filling structure is higher than a top of the planarization layer.
6. The display device of claim 1, wherein the definition layer further comprises a first spacer extending upward in such a way that a top of the first spacer is higher than a top of the covering portion, and the first spacer is located in the first region.
7. The display device of claim 6, wherein at least a portion of the first spacer does not overlap the first electrodes in plan view.
8. The display device of claim 6, wherein the definition layer further comprises a second spacer extending upward in such a way that a top of the second spacer is higher than the top of the covering portion, and the second spacer is located in the second region.
9. The display device of claim 8, wherein a height of the first spacer is the same as a height of the second spacer.
10. The display device of claim 8, wherein an area of the first spacer is greater than an area of the second spacer in plan view.
11. The display device of claim 8, wherein the second electrode overlaps the first and second spacers.
12. The display device of claim 8, wherein the lower structure further comprises a charge transport layer located between the first and second electrodes, and the charge transport layer overlaps the first and second spacers.
13. The display device of claim 8, wherein the lower structure further comprises an organic passivation layer located directly on the second electrode, and the organic passivation layer

overlaps the first and second spacers.

14. The display device of claim 1, wherein the substrate comprises the plastic, and the inorganic multilayer expands to completely cover the first region in plan view.

15. The display device of claim 14, wherein the substrate has a recess formed at a lower portion of the substrate to correspond to the first region.

16. The display device of claim 1, wherein the substrate comprises the glass, and the substrate has a recess formed at a lower portion of the substrate to correspond to the first region.

17. The display device of claim 1, wherein the substrate has a hole corresponding to the first region, and the inorganic multilayer expands to completely cover the first region in plan view.

18. The display device of claim 1, wherein the plurality of first display units comprises an adjacent display unit located adjacently to the first region, and the adjacent display unit always has a black color.

19. The display device of claim 1, wherein the plurality of first display units comprises an adjacent display unit located adjacently to the first region, and the adjacent display unit has a black color when an optical unit located under the display device to correspond to the first region is used.

20. The display device of claim 1, wherein the lower structure further comprises a charge transport layer located between the first and second electrodes, and the charge transport layer has a hole corresponding to the first region.

21. The display device of claim 1, wherein the second electrode has a hole corresponding to the first region.

22. The display device of claim 21, wherein the second electrode is transreflective.

23. The display device of claim 21, wherein the hole of the second electrode is at least partially filled with a filling structure comprising a transparent resin.

24. The display device of claim 1, wherein the lower structure further comprises an organic passivation layer located directly on the second electrode, and the organic passivation layer has a hole corresponding to the first region.

25. The display device of claim 24, wherein the hole of the organic passivation layer is at least partially filled with a filling structure comprising a transparent resin.

26. The display device of claim 1, wherein the lower structure further comprises an organic passivation layer located directly on the second electrode, and a stack comprising the second electrode and the organic passivation layer has a hole corresponding to the first region.

27. The display device of claim 26, wherein the hole of the stack comprising the second electrode and the organic passivation layer is at least partially filled with a filling structure comprising a transparent resin.

28. The display device of claim 1, further comprising a color filter located on the upper encapsulator, wherein the color filter has a hole corresponding to the first region.

29. The display device of claim 28, wherein the color filter comprises color filtering regions and a light blocking region.

30. The display device of claim 29, wherein the light blocking region comprises a black matrix.

31. The display device of claim 28, wherein the definition layer has a color.

32. The display device of claim 31, wherein the color filter is used instead of a polarization layer.

33. The display device of claim 1, further comprising a polarization layer located on the upper encapsulator, wherein the polarization layer has a hole corresponding to the first region.

34. The display device of claim 1, wherein the first electrodes are not located to correspond to the first region.

35. The display device of claim 1, wherein the emission layers are not located to correspond to the first region.

36. The display device of claim 1, further comprising touch electrodes located on the upper encapsulator, wherein the touch electrodes are not located to correspond to the first region.

37. The display device of claim 1, wherein the definition layer has a hole corresponding to the first

region.

38. The display device of claim 37, wherein the definition layer has a color, and the hole of the definition layer is at least partially filled with a filling structure comprising a transparent resin.

39. The display device of claim 37, wherein the planarization layer has a hole corresponding to the first region, the definition layer and the planarization layer are colored, and the hole of the definition layer and the hole of the planarization layer are at least partially filled with a filling structure comprising a transparent resin.

40. The display device of claim 1, wherein the planarization layer has a hole corresponding to the first region.

41. The display device of claim 40, wherein the planarization layer has a color, and the hole of the planarization layer is at least partially filled with a filling structure comprising a transparent resin.

42. The display device of claim 1, wherein the planarization layer has a first portion corresponding to the first region and a second portion corresponding to the second region, and a bottom of the first portion is lower than a bottom of the second portion.

43. The display device of claim 1, wherein the upper encapsulator comprises an inorganic layer.

44. The display device of claim 1, further comprising a filling structure corresponding to the first region, wherein the filling structure comprises a transparent resin, a top of the filling structure is higher than a top of the planarization layer, and a bottom of the filling structure is lower than the top of the planarization layer.
